Circuit Analysis (Lab10)

Experiment 9: Verification of Thevenin's Theorem

Name: TALHA YOUSAF **Registration No: 2025-CE-93**

Date: 5/11/2026 **Grade and Signature: ……**

CLO2 **CLO3**	Use with identification the circuit components, breadboards, multimeter, power supplies, signal generators, and oscilloscopes. Make measurements in their own constructed electric circuits.			
Psychomotor/Affective	**Level1 (1)**	**Level2 (2-3)**	**Level3 (4-5)**	**Level4 (6-7)**
Report Marks (3)				
	Total marks (10)			

Objective

The objective of this lab is verify Thevenin's theorem and to find the full load current.

Apparatus Required

Sr. no	Apparatus	Range	Quantity
1	Regulated Power Supply	Standard	1
2	Signal Generator	Standard	1
3	Oscilloscope	Standard	1
4	Digital Multimeter	Standard	1
5	Breadboard & Wires	Standard	As Required
6	Resistors	1k, 330 ohms	1,3
7	Potentiometer	Standard	1

Description

Statement:

Any linear bilateral, active two terminal network can be replaced by an equivalent voltage source (VTH). Thevenin's voltage or V OC in series with looking pack resistance RTH.

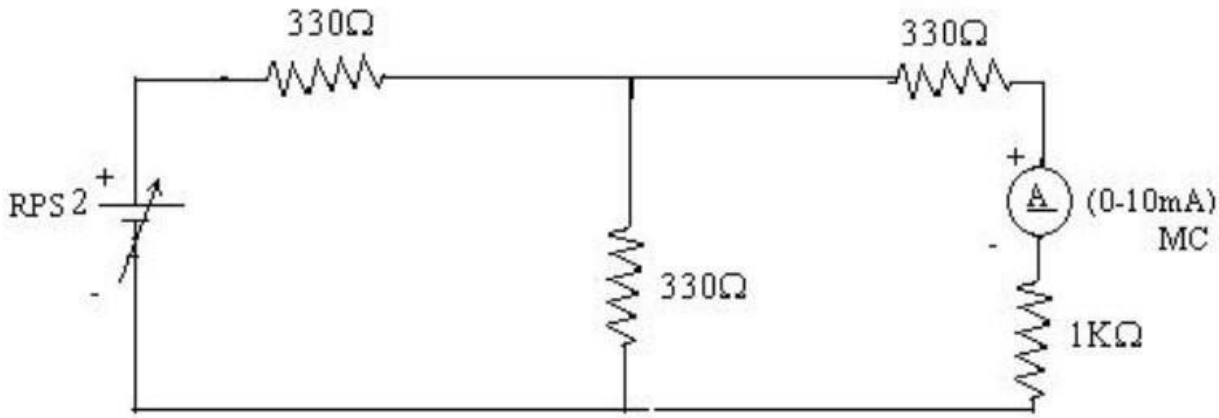
Precautions:

1. Voltage control knob of RPS should be kept at minimum position.
2. Current control knob of RPS should be kept at maximum position

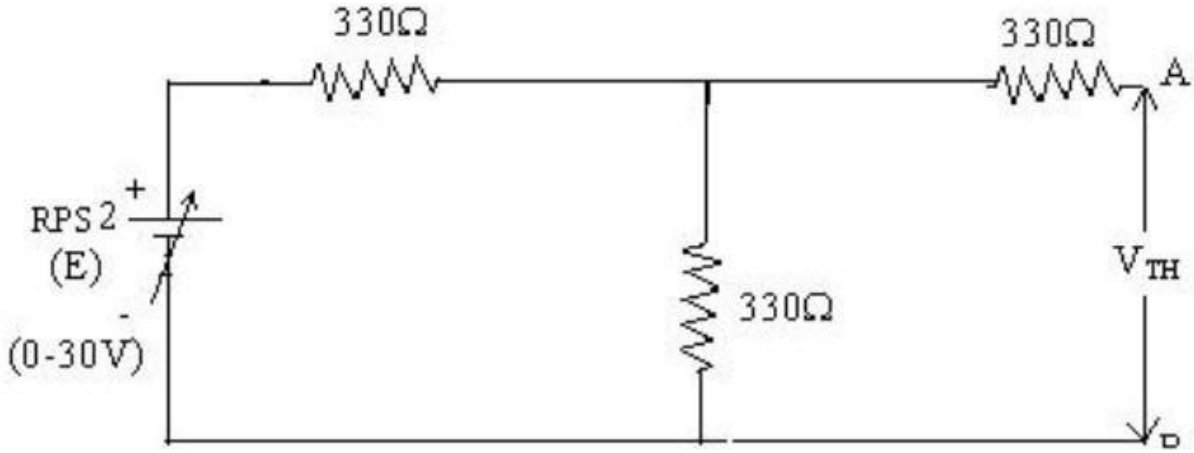
Procedure:

1. Connections are given as per the circuit diagram.
2. Set a particular value of voltage using RPS and note down the corresponding ammeter readings. To find VTH 3. Remove the load resistance and measure the open circuit voltage using multimeter (VTH). To find RTH
4. To find the Thevenin's resistance, remove the RPS and short circuit it and find the RTH using multimeter.
5. Give the connections for equivalent circuit and set VTH and RTH and note the corresponding ammeter reading. 6. Verify Thevenin's theorem.

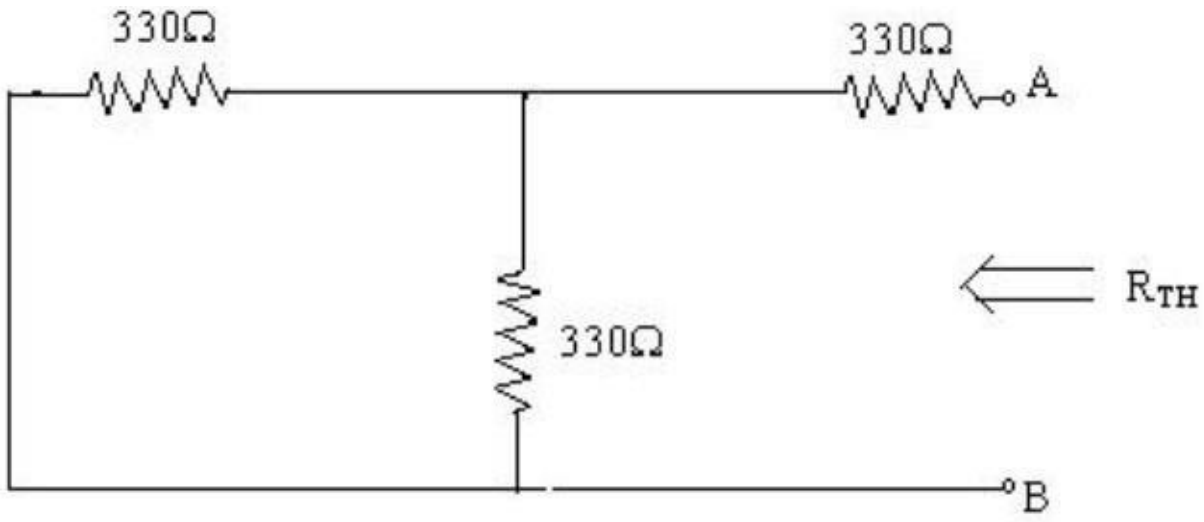
Circuit - 1: To find load current



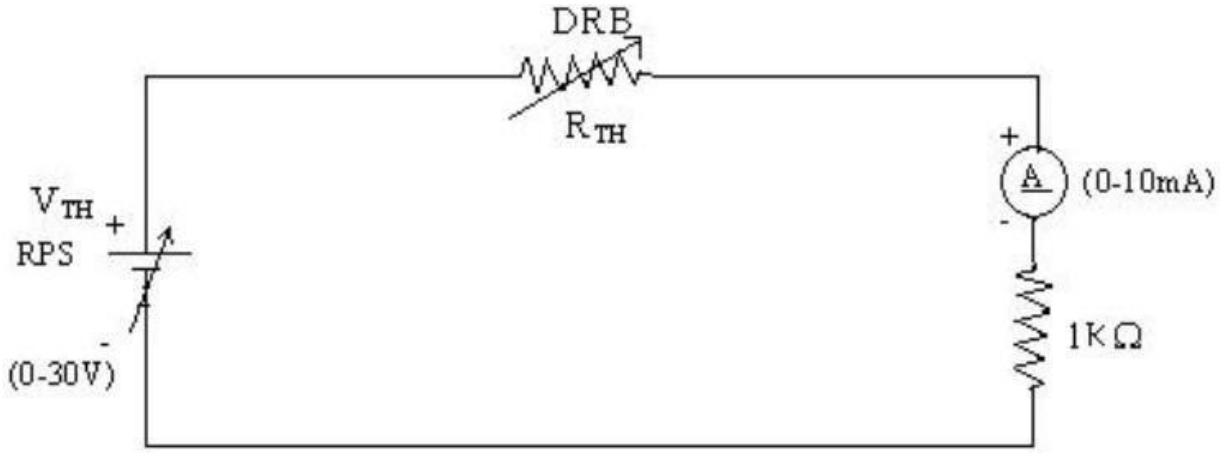
To find V_{TH}



To find R_{TH}



Thevenin's Equivalent circuit:



Thevenin Practical Values

	E(V)	VTH(V)	RTH(ohms)	I_L (mA)	
				Circuit - I	Equivalent Circuit
Values	**25V**	**12.5V**	**495 Ohms**	**8.36mA**	**8.36 mA**

Post Lab Questions

1. What is Thevenin's Theorem?

Thevenin's Theorem states that any linear, bilateral network containing one or more voltage sources and resistances can be replaced by an equivalent circuit consisting of a single voltage source Vth in series with a single resistance Rth. The theorem focuses on the behavior of the circuit from the perspective of two output terminals (the load), simplifying complex networks into a manageable two-component model.

---

2. **What are the applications of Thevenin's Theorem?**

- Circuit Simplification: It allows engineers to reduce a large, complex circuit into a simple series circuit, making it much easier to analyze.
- Load Analysis: It is particularly useful when the load resistance RL changes frequently. Instead of re-calculating the entire circuit,
- Power System Analysis: It is used in power grids to determine the short-circuit current and to analyze how adding new components will affect the overall system stability.
- Maximum Power Transfer: It provides the foundation for the Maximum Power Transfer Theorem, which states that maximum power is delivered to a load when

---

3. **What are the theoretical and practical limitations?**

Theoretical Limitations

- Linearity: The theorem only applies to linear circuits. It cannot be used for circuits containing non-linear components like diodes or transistors (unless analyzed over a small linear range).
- Bilateral Components: It only works for networks where the components have the same properties in both directions (e.g., resistors, inductors, capacitors).
- Dependent Sources: While it can handle circuits with dependent sources, the calculation becomes significantly more complex, as you cannot simply "short" a dependent source to find Rth

Practical Limitations

- Internal Resistance: In a real lab setting, the voltage source and measuring instruments (ammeters/voltmeters) have their own internal resistances which can cause slight discrepancies between theoretical and practical V_{TH} values.
- Power Dissipation: A Thevenin equivalent circuit is only equivalent at the terminals. It does not represent the internal power dissipation or efficiency of the original complex circuit; the original circuit might be wasting energy as heat that the simplified model doesn't show.